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## CO2 - AT3 - CLASS TEST

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Course code: ITAD518

Course: Computer vision.

I Q & A

1) A dark image lacks details.

- \* The best gray level transformation is log transformation.
- \* It increases the brightness of dark pixels while keeping bright pixels almost unchanged.
- \* This makes hidden details in dark regions visible.
- \* Applications: Apply the log function to each pixel value.
- \* Result: The image becomes brighter and details in low-intensity areas are clearly visible.

2) An image has poor contrast.

- \* The suitable technique is histogram equalization.
- \* It redistributes pixel intensity values over the full range.
- \* This improves the contrast of the image.
- \* Application: compute the histogram and equalize the intensity values.
- \* Result: The image looks like clearer, and objects become easier to distinguish.

3) A noisy image contains random intensity variations.

- \* The best smoothing filter is the Mean filter.
- \* It replaces each pixel with the average of its neighbours.
- \* This reduces random noise.

\*> Result: The image becomes smoother, but very fine details may be slightly blurred.

4) An object is clearly brighter than the background.

\*> The suitable segmentation technique is Thresholding.

\*> A threshold value separates the object from the background.

\*> Result: The object is extracted quickly and accurately.

5) An image contains salt-and-pepper noise.

\*> The best filter is the Median filter.

\*> It replaces each pixel with the median value of its neighbouring pixels.

\*> Result: Noise is removed while important image details are preserved.